

Tribological studies of polyamide 6 and high-density polyethylene blends filled with PTFE and copper oxide and reinforced with short glass fibers

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Abstract

The mechanical and tribological behaviors of polyamide 6 (PA6) and high density polyethylene (HDPE) polyblends made using maleic anhydride polypropylene as the compatibilizing agent were studied. The compositions investigated for tribological behavior were 80 wt.% PA6–20 wt.% HDPE and 60 wt.% PA6–40 wt.% HDPE. The polyblends were reinforced with glass fiber (GF) and filled with polytetrafluoroethylene (PTFE) and copper oxide (CuO). The polymeric materials were blended using a modular intermeshing co-rotating twin-screw extruder. The reinforced composite specimens with different fiber proportions were made in a reciprocating screw injection molding machine using the blended materials as the feed stock. The friction and wear experiments were run under ambient conditions in a pin-on-disk machine with the polymer pin riding on the flat surface of a steel disk at a sliding speed of 1 ms^{-1} and under a nominal pressure of 0.64 MPa. The tensile strength of the polyblend with composition 80 wt.% PA6–20 wt.% HDPE increased with fiber reinforcement but the material became brittle. The maximum reductions in wear and the coefficient of friction were obtained by filling the polyblends with 10 wt.% PTFE. The fiber glass reinforcement was not found effective in lowering the already low wear rate of the polyblend.

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1. Introduction

Composite materials find applications for a variety of parts because they offer the combinations of properties which cannot be attained with metals, ceramics or polymers alone. Polymer composites have, therefore, replaced many traditional metallic materials for sliding components. The increasing use of polymer composites in sliding components has made the studies of the tribological behavior of these materials a commercial necessity. For the development of new materials, polyblending offers an alternative to copolymerization. The newly developed materials may be tailored in terms of composition so as to have the reasonable mix of properties of the polymers used for polyblending. The properties may be improved further by reinforcements and fillers [1], resulting in higher strength, high wear resistance, low coefficient of friction, etc. The properties of such a composite would be affected by the distribution of polymer phases, the compatibilizer used for adhesion between the phases, and the type of reinforcement or filler [2].

The polyblends of polyamide 6 (PA6) and high density polyethylene (HDPE) are attractive because of the desirable properties of each polymer. PA6 exhibits a high crystalline melting point, good resistance to solvents, high strength, ease of fabrication and processing, and high wear resistance. HDPE has excellent low temperature flexibility, low cost, resistance to moisture permeation, and good tribological properties. Therefore, the blends of PA6 and HDPE may be prepared to provide the most desirable properties of both polymers, while avoiding their major drawbacks. In view of these considerations, the authors reported in an earlier study [3] the structure, mechanical properties, and tribological behaviors of the polyblends made from these polymers. The latter were made both with and without the compatibilizer. This study revealed that the two polyblend compositions, 60 wt.% PA6–40 wt.% HDPE and 80 wt.% PA6–20 wt.% HDPE, prepared using the compatibilizer had the highest wear resistance and very low coefficient of friction. With the aim of improving the properties of these polyblends further, it was decided in this work to reinforce the polyblend with short glass fibers (GF) and, alternatively, to fill these with the organic filler polytetrafluoroethylene (PTFE) and inorganic filler copper oxide (CuO).

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The mechanisms of wear reported in sliding of filled and reinforced polymer composites against metal counterfaces are adhesion and abrasion. An important characteristic of adhesive wear is the transfer of soft polymeric material to the harder metal counterface [4]. It has been found that some fillers and fibers are very effective in reducing wear under adhesive wear conditions [5–9]. On the other hand, in situations where abrasive wear is involved, fibers may hurt the wear resistance of the polymer. Various studies [6–9] have demonstrated that the wear resistance of polymers is improved by the addition of fillers. Some of the fillers that are effective in reducing wear are CuO, CuS, CuF₂, PbS, and CaS. Bahadur et al. [7,10] demonstrated that the decomposition and physicochemical reactions of the fillers with the counterface contributed to the formation of a thin, stable and adherent transfer film resulting in increased wear resistance. It has been known for quite some time that PTFE polymer exhibits a self-lubricating behavior and its application in sliding prevents stick-slip motion instabilities. Despite its low-friction properties, PTFE wears excessively at typical engineering speeds. It is thus used as an additive in wear-resistant polymers. The studies on the friction and wear of polymer composites containing PTFE have reported that PTFE reduces the coefficient of friction and wear rate of some composites but increases the wear rate of others [9]. This makes it imperative to evaluate the effect of its addition to new materials such as the polyblends investigated in this work.

The addition of reinforcing agents such as glass and carbon, particularly in the form of fibers, enhances the mechanical properties of polymers. This is generally good for tribological properties as well. While unidirectional fiber reinforcements are more effective in this respect, their processing is rather expensive. On the other hand, short fiber-reinforced polymers can be processed cheaply but are not that effective. The reinforcement with fibers has been reported to improve the wear resistance of PA, PEEK and polyethersulfone [5,11]. Bahadur and Zheng [12] found that 10 wt.% short fiber glass reinforcement in polyester provided the optimum wear resistance.

Based on the above observations and in our effort to develop highly wear-resistant composites, it was decided to fill the two HDPE–PA6 polyblend compositions found to have excellent wear resistance [3] with inorganic and organic fillers and also to reinforce with short GFs.

2. Experimental details

Polyblends of PA6 and HDPE were prepared using maleic anhydride modified polypropylene (MAGPP) as the compatibilizer. The polyblends were produced in two compositions, 60 wt.% PA6–40 wt.% HDPE and 80 wt.% PA6–20 wt.% HDPE. They were found to have the highest wear resistance along with very low coefficient of friction in our earlier study [3]. The amount of compatibilizer added was 5 wt.% based on the recommendation by Lim and White [13] who reported that this compatibilizer proportion was adequate for interaction between PA6 and HDPE interfaces. The polyblends were filled with PTFE in 5 and 10 wt.% proportions. The two polyblends with 10 wt.% PTFE were further filled with 30 vol.% (69 wt.%) CuO. These proportions were selected based on earlier studies [7,14] which have shown that these are the most effective proportions for reduction of wear and the coefficient of friction. The polyblends were also reinforced with GF in proportions of 5, 10 and 15 wt.%. The fibers were 17 μ m diameter and 6 mm long. The sources and the characteristics of these materials are listed in Table 1.

The polymers, fillers and compatibilizer were dried at 85 °C for 48 h to avoid plasticization and hydrolyzing effects from humidity. The materials were mixed and the mixture was extruded using an 18 modular intermeshing co-rotating twin-screw extruder. The temperatures maintained in five zones of the extruder barrel were 180, 235, 240, 245, and 240 °C and the temperature at the die was 235 °C. The extruder screw speed was set at 80 rpm which yielded a feed rate of 6 kg/h. The extrudate was obtained in the form of a cylindrical rod which was quenched in cold water and then pelletized. The initial extruded material was discarded to flush impurities out of the system before getting the polyblend samples.

The specimens for tensile tests (ASTM D638, Type 1) and some tribological tests were injection molded from the pelletized polyblend material using a reciprocating screw injection molding machine. The GF for reinforcement was added during the injection molding process. The temperatures maintained in the three zones of the molding machine barrel were 215, 220, and 240 °C, in the nozzle 230 °C, and in the mold 90 °C. The screw speed was set at 10 rpm and the injection pressure at 800 bar. The times for injection, cooling, and ejection were 8, 30, and 1.5 s, respectively. The

Table 1
Properties of materials used for polyblends, fillers and reinforcement

Material	Designation	Density (kg/m ³)	Melting point (°C)
Polyamide 6	PA6	1130	213
High density polyethylene	HDPE	960	137
Polytetrafluoroethylene	PTFE	2160	327
Maleic anhydride polypropylene	MAGPP	940	133
Glass fiber	GF	2500	–
Copper oxide	CuO	6300	–

molded specimens were visually inspected for air bubbles and those with defects were discarded.

Friction and wear tests were performed in a pin-on-disk machine in which the stationary polymer pin was in contact with a rotating steel disk. The size of polymer pin was 6 mm diameter and 18 mm long. Before testing, the pins were abraded against a 600-grade SiC emery paper to ensure a good contact between the sliding pin surface and the counterface during the sliding experiment. The counterface was made of AISI 02 tool steel (0.90% C, 1.6% Mn) which was quench-hardened. The size of rotating disk was 75 mm diameter and 5 mm thick. It was hardened to HRC 58 and finished by grinding followed by polishing with 320-grade SiC emery paper to give a surface roughness of 0.10–0.11 μm R_a . Dry sliding at ambient condition (with 35–40% relative humidity) with a normal load of 19.6 N (0.64 MPa nominal pressure) and a sliding speed of 1 ms^{-1} was performed in a track of 65 mm diameter. Before testing, the pin and the disk surfaces were cleaned with acetone and thoroughly dried.

The amount of wear was measured by interrupting the sliding test at suitable intervals for weighing the pin to an accuracy of 0.0001 g in a precision balance. The friction force was measured from the output of a pair of strain gauges mounted on the vertical arm that carried the pin. For each condition, at least three tests were performed and the plotted values are the means calculated from the data of these tests. The variation in the coefficient of friction was about $\pm 10\%$ and in wear $\pm 20\%$ from test to test. The transfer behavior of polymeric materials to the counterfaces in rubbing was studied by optical microscopy.

3. Results and discussion

Blending PA6 with HDPE results in an immiscible two-phase blend. The latter is characterized by a coarse, meta-stable morphology, and poor adhesion between the phases. This gives rise to a phase-separated morphology which is not good for processing or properties. It was shown in our earlier paper [3] that blending with compatibilizer led to fusion between the two phases. MAgPP was used as the compatibilizer because polypropylene is miscible with other polyolefins and the grafted anhydride chain is expected to establish bonding with polyamide.

3.1. Mechanical properties

The effect of PTFE filler and fiber reinforcement on the mechanical properties was studied on the polyblend with composition 80 wt.% PA6–20 wt.% HDPE only. The tensile tests were performed at the crosshead speed of 5 mm/min. When PTFE was added to the polyblend in two proportions of 5 and 10 wt.%, there was practically no effect on the tensile strength of the polyblends, as seen from Fig. 1. When the polyblend was reinforced with short GF, there was 20% increase in tensile strength with as little as 5% GF. With

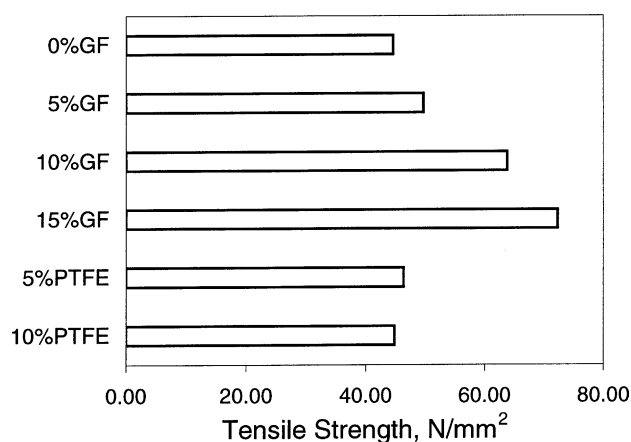


Fig. 1. Tensile strength of 80 wt.% PA–20 wt.% HDPE polyblend filled with PTFE and reinforced with GF in the proportions stated.

the addition of 10% GF, the tensile strength increased from 45 to 63.5 N/mm^2 , and with 15% GF it increased further to 72.5 N/mm^2 which is a 60% increase. As for the hardness (Fig. 2), there was a slight increase in hardness with glass reinforcement and the effect of PTFE addition was negligibly small.

With the addition of PTFE, the ductility of the polyblend decreased slightly (Fig. 3). When the material was reinforced with GFs in any of the three proportions stated earlier, the reinforced polyblends behaved as brittle materials. Thus, the only effect to be noted here is that GF reinforcement increases the tensile strength but makes the material brittle. The effect of PTFE filler on tensile strength, hardness and elongation-to-break is fairly small.

3.2. Wear and friction

HDPE sliding at 1 ms^{-1} provided the coefficient of friction of 0.31 but a high steady-state wear rate of 4.33 $\times$

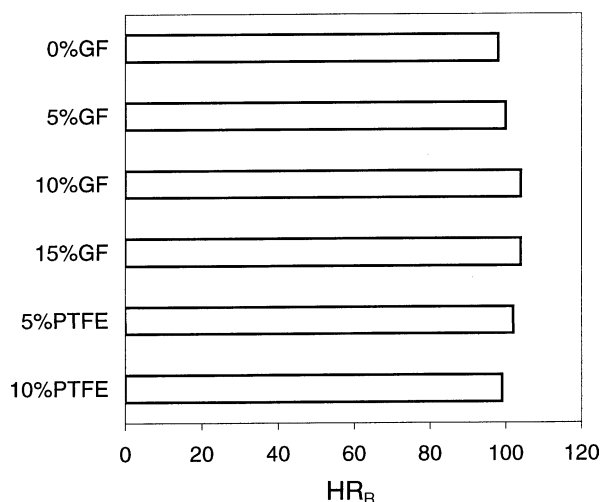


Fig. 2. Rockwell hardness (HR_R) of 80 wt.% PA–20 wt.% HDPE polyblend filled with PTFE and reinforced with GF in the proportions stated.

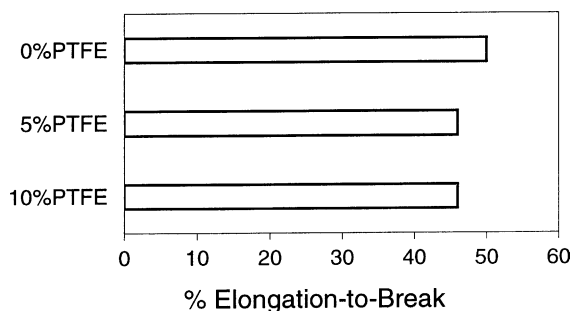
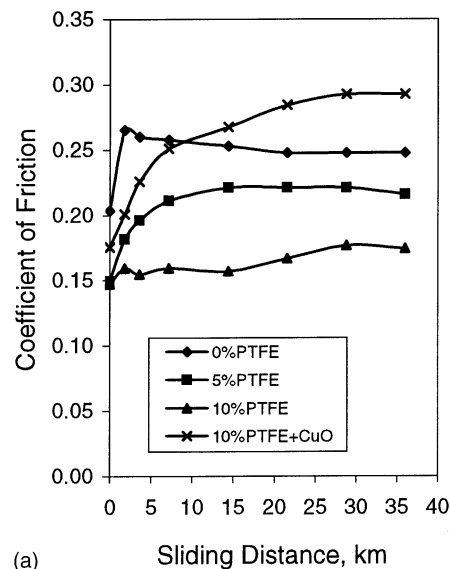


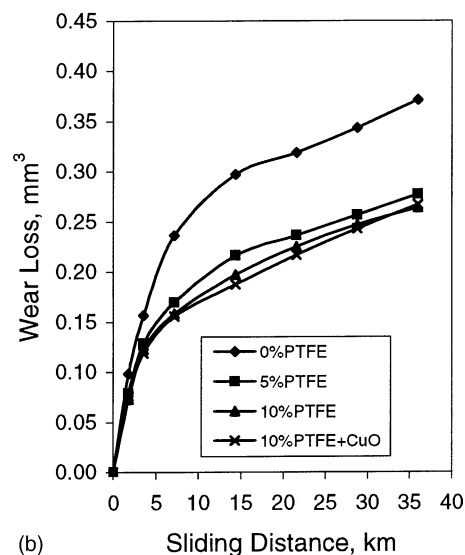
Fig. 3. Effect of PTFE filler on the % elongation-to-break of 80 wt.% PA–20 wt.% HDPE.

$10^{-6} \text{ mm}^3/\text{Nm}$. On the other hand, PA6 had a much higher coefficient of friction of 0.82 but a lower wear rate of $2.80 \times 10^{-6} \text{ mm}^3/\text{Nm}$. None of the two polymers is a good choice for the polymer–metal sliding combination. On the other hand, when the polyblends of these two polymers with 60 and 80 wt.% PA6 were tested, the friction and wear behaviors were remarkably better, as seen in Table 2. In an effort to make further improvements in the tribological behavior of these polyblends, the three possible approaches were tried. These included filling the polyblends with organic material PTFE, inorganic material CuO, and reinforcing with GF. The results of these approaches are given below.

Fig. 4(a) shows the variation of the coefficient of friction with sliding distance for 80 wt.% PA6–20 wt.% HDPE polyblend for two proportions of PTFE. There is a slight reduction in the coefficient of friction with 5% PTFE, but the reduction is substantial when 10% PTFE is added. The coefficient of friction obtained in this case is 0.15–0.17 which is remarkably low for dry sliding at 1 ms^{-1} considering that the coefficient of friction of PTFE at this sliding speed is



(a)



(b)

Fig. 4. Variation of: (a) the coefficient of friction and (b) wear volume loss with sliding distance for 80 wt.% PA–20 wt.% HDPE polyblend filled with PTFE, and PTFE and CuO. Test conditions: 19.6 N load (0.64 MPa normal pressure); 1.0 ms^{-1} sliding speed.

about 0.3 [15]. The effect of PTFE addition on wear exhibits in two ways. One of them is that steady-state wear commences earlier so that the wear volume for any sliding distance is lower than that without PTFE. The second point is that the steady-state wear rate is reduced from 0.22×10^{-6} to $0.18 \times 10^{-6} \text{ mm}^3/\text{Nm}$ (see Table 2).

When the polyblend with composition 60 wt.% PA6–40 wt.% HDPE was filled with 5 and 10 wt.% PTFE, there was a slight reduction in the coefficient of friction but a much larger reduction in wear, as seen in Fig. 5 and Table 2. The wear rate of $0.13 \times 10^{-6} \text{ mm}^3/\text{Nm}$ for this polyblend with PTFE addition was even lower than that of the other polyblend with PTFE added. Since the coefficient of friction of this polyblend without PTFE was lower than that of

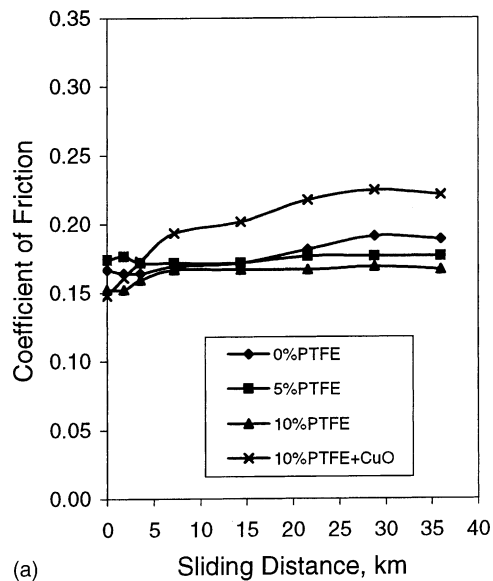
Table 2

Coefficients of friction and steady-state wear rates for polymers and their polyblends in the filled and reinforced conditions

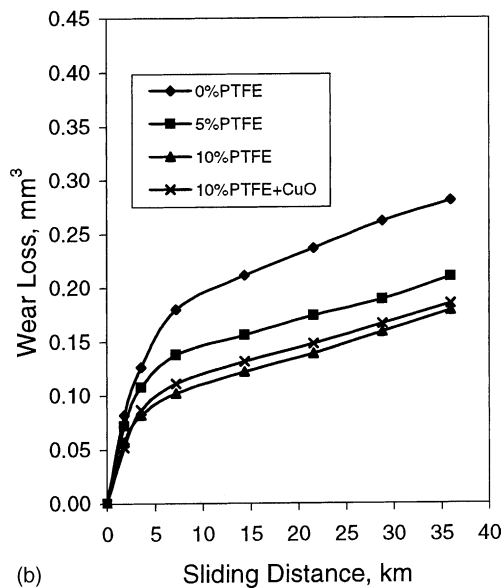
Material	Coefficient of friction	Wear rate ($\times 10^{-6} \text{ mm}^3/\text{Nm}$)
HDPE	0.31	4.33 [3]
PA6	0.82	2.80 [3]
(a) 80 wt.% PA6–20 wt.% HDPE	0.25	0.22 ^a [3]
5 wt.% PTFE	0.22	0.18 ^a
10 wt.% PTFE	0.17	0.18 ^a
10 wt.% PTFE and CuO ^b	0.29	0.20
5 wt.% GF	0.24	0.26
10 wt.% GF	0.24	0.25
15 wt.% GF	0.27	0.35 ^a
(b) 60 wt.% PA6–40 wt.% HDPE	0.19	0.18 [3]
5 wt.% PTFE	0.18	0.13
10 wt.% PTFE	0.17	0.13
10 wt.% PTFE and CuO ^b	0.22	0.13

^a Wear rates were calculated by regression analysis from the last four data points in Figs. 4 and 6. In other cases, the data points used for regression analysis were five.

^b These compositions represent polyblends with 10 wt.% PTFE which were further filled with 30 vol.% (69 wt.%) CuO.



(a)

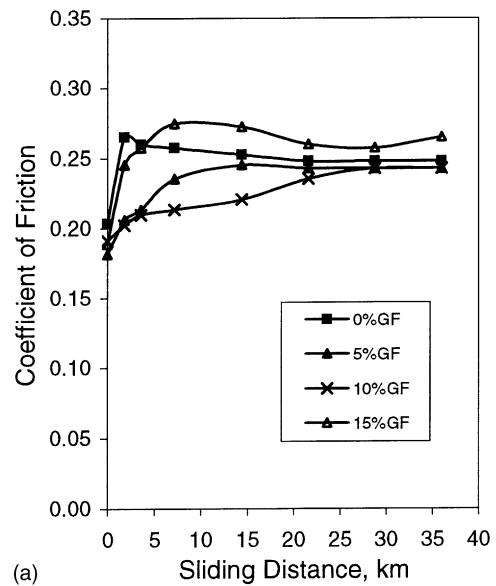


(b)

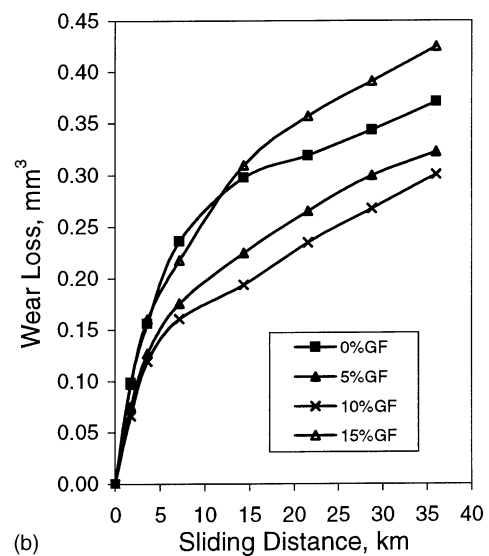
Fig. 5. Variation of: (a) the coefficient of friction and (b) wear volume loss with sliding distance for 60 wt.% PA–40 wt.% HDPE polyblend filled with PTFE, and PTFE and CuO. Sliding conditions same as in Fig. 4.

80% PA6–20% HDPE, the reduction in the coefficient of friction with PTFE from 0.19 to 0.17 was smaller. Thus, the coefficient of friction of 0.17 seems to be about the lowest value for dry sliding.

Fig. 4 shows also the friction and wear behaviors when 80 wt.% PA6–20 wt.% HDPE polyblend with 10 wt.% PTFE was filled with CuO. The most significant effect is that the coefficient of friction increased from 0.22 to 0.29. There is insignificant effect on steady-state wear rate. The behaviors in respect of both the coefficient of friction and wear rate were identical when CuO was added to 60 wt.% PA6–40 wt.% HDPE polyblend with 10 wt.% PTFE (see Fig. 5, Table 2).



(a)



(b)

Fig. 6. Variation of the coefficient of friction and wear volume loss with sliding distance for 80 wt.% PA–20 wt.% HDPE polyblend reinforced with 5, 10 and 15 wt.% GF. Sliding conditions same as in Fig. 4.

The results of the coefficient of friction and wear for 80 wt.% PA6–20 wt.% HDPE when reinforced with GF are shown in Fig. 6. Whereas some differences in the coefficient of friction are noticed in the early part of sliding, the differences become insignificant later on. With 5 and 10 wt.% GF, steady-state wear commences earlier than with no reinforcement but there is no significant effect on wear rate. Normally, the fiber reinforcement in 10 wt.% proportion has been reported to reduce wear [12]. It is suspected that the wear reduction here did not occur because the steady-state wear rate of the non-reinforced polyblend itself was fairly small. With 15% GF reinforcement, the transient wear state became longer and the steady-state wear rate increased (Table 2).

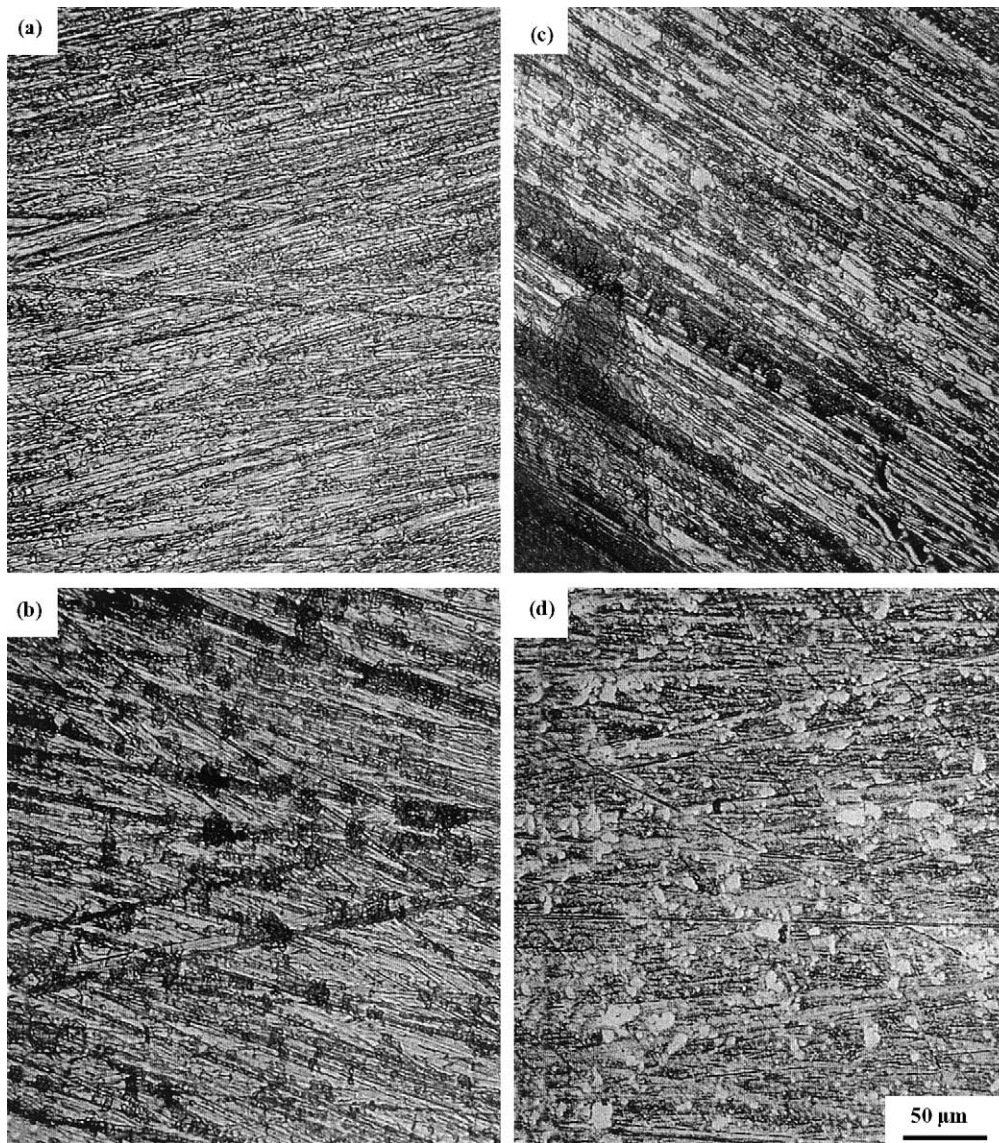


Fig. 7. Optical micrographs of the steady-state transfer films formed by 80 wt.% PA6–20 wt.% HDPE polyblends with the compositions stated: (a) no PTFE and no GF; (b) with 10 wt.% PTFE; (c) with 10 wt.% PTFE and CuO; (d) reinforced with 15 wt.% GF. Same magnification as marked in 7(d).

3.3. Transfer films

Fig. 7 shows the micrographs of wear tracks of the steel surfaces that were involved in sliding against 80 wt.% PA6–20 wt.% HDPE with different filler and reinforcement proportions. From the micrograph in Fig. 7(a), it is seen that the polyblend with no filler and no reinforcement forms a good transfer film. The film is uniform and thin, and the deep scratches left over from the finishing operation are barely seen. Since the film provides protection to the pin surface from the aggressive damage of hard metal asperities, the wear rate of this composition was small (Table 2).

The transfer films were also formed in the case of polyblends filled with 5 and 10 wt.% PTFE, as shown in Fig. 7(b). Since good transfer film is formed with this material even in the unfilled condition, the role of protection from trans-

fer film is not governing the wear behavior in this cases. It is probably the nature of transfer film that is responsible for some reduction in wear, as PTFE has been reported to form a molecularly smooth transfer film with low shear strength [6]. The same was the case with the polyblend with 10 wt.% PTFE and CuO as well, as it also formed a good transfer film (Fig. 7(c)).

The micrograph in Fig. 7(d) shows that transfer films is formed on the counterface even when the polyblend is filled with GF. The shiny spots represent the presence of GF in transfer material. There appears to be same disruption of transfer film from GF and this could be responsible for a slight increase in wear rate.

Similar transfer films were observed in the case of 60 wt.% PA6–40 wt.% HDPE polyblend with different filler proportions as well. These are shown for some cases in Fig. 8.

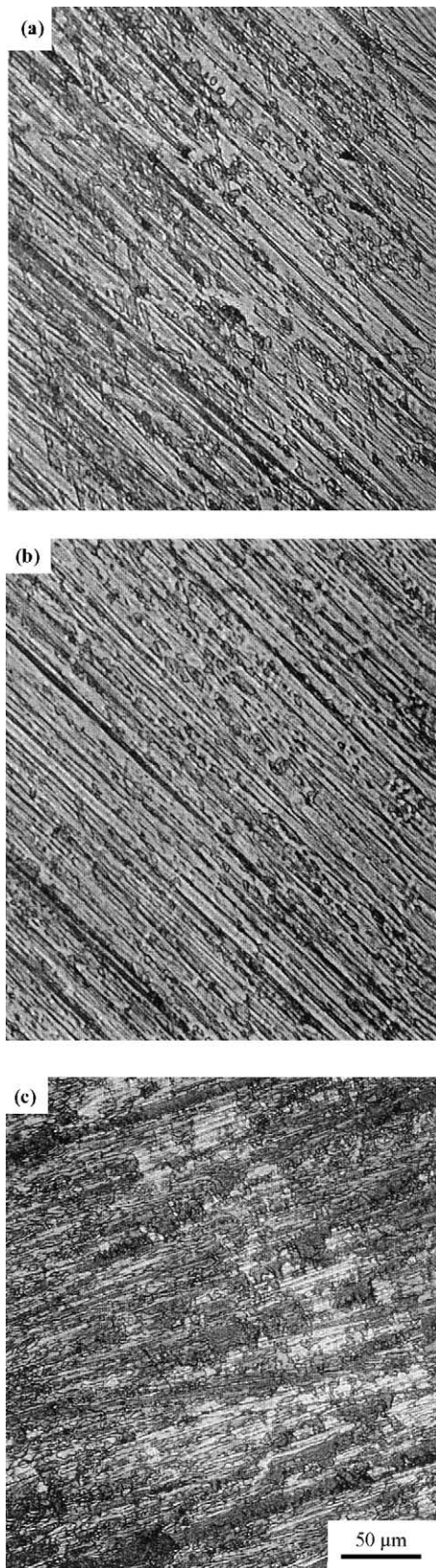


Fig. 8. Optical micrographs of the steady-state transfer films formed by 60 wt.% PA6–40 wt.% HDPE polyblends with the compositions stated: (a) unfilled; (b) filled with 10 wt.% PTFE; (c) filled with 10 wt.% PTFE and CuO. Same magnification as marked in (c).

3.4. Discussion

The polyblends selected for studies in this paper had very good tribological properties to begin with. They also formed good transfer films on the counterfaces during sliding which is generally a prerequisite for good tribological properties. It was thus a challenge to improve their tribological properties further. The most common approach tried in the literature is by reinforcing the polymer material with fibers. Consequently, the polyblends were reinforced with three proportions of GF. As a result of it, the tensile strength of the materials increased but no significant reduction in wear or coefficient of friction was observed. Instead, the wear rate increased with 15 wt.% GF which is consistent with other studies [12]. These observations clearly indicate that the increase in tensile strength does not necessarily contribute to a higher wear resistance.

Consistent with the results reported elsewhere [7,14], the polyblends were filled with 10 wt.% PTFE and 30 vol.% CuO, because 20–30 vol.% CuO has been observed to reduce the wear rate of many polymers. The adverse effect of the particulate inorganic filler reported in the literature is the increase in the coefficient of friction [7]. That is why CuO was added to the polyblend with 10 wt.% PTFE. The coefficient of friction was still higher by about 15% than that of the unfilled polyblends. The transfer film which developed in sliding of this material did not make a difference because the unfilled polyblend itself formed a good transfer film. Considering that the particulate fillers do not increase the strength by any significant amount, there would be no advantage in using the polyblend with this filler combination.

PTFE as the filler in both 80 wt.% PA6–20 wt.% HDPE and 60 wt.% PA6–40 wt.% HDPE polyblends reduced steady-state wear rate and also the coefficient of friction in some cases. The maximum reduction in steady-state wear rate was 28% and in the coefficient of friction 10% with 10 wt.% PTFE in 60 wt.% PA6–40 wt.% HDPE. The corresponding reductions in the other polyblend 80 wt.% PA6–20 wt.% HDPE were 10 and 32%, respectively. The polyblend 60 wt.% PA6–40 wt.% HDPE with 10 wt.% PTFE had the lowest wear rate and also the lowest coefficient of friction. Thus, this is the most desirable composition for tribological applications. However, if strength is also important along with tribological behavior, the polyblend with 80 wt.% PA6–20 wt.% HDPE filled with 10 wt.% PTFE could be a better choice because its tensile strength is 45 N/mm² compared to 31 N/mm² for the other polyblend with the same PTFE content (as determined from tests).

Since transfer films were formed in all cases, the mechanism of wear was the same. It consisted of the development of transfer film on the counterface during transient wear state through the process of microfragmentation of polymeric material because of interaction with metal asperities. Once the transfer film was formed on the counterface, wear was contributed by the loss of transfer film fragments during repetitive sliding and occasional separation of material from the

pin surface and escaping from the interface as opposed to building up the transfer film.

4. Conclusions

1. When 80 wt.% PA6–20 wt.% HDPE polyblend was reinforced with 5–15 wt.% GF, the tensile strength of the polyblend increased with increasing fiber content but the material became brittle. There was, however, no significant effect on the tensile strength, hardness and elongation-to-break with the addition of PTFE to the polyblends.
2. With the additions of PTFE in two proportions, the steady-state coefficients of friction of 80 wt.% PA6–20 wt.% HDPE decreased significantly while the reductions in case of 60 wt.% PA6–40 wt.% HDPE were smaller. Furthermore, the steady-state wear commenced earlier than with no PTFE and the steady-state wear rate decreased significantly.
3. There was no advantage seen in terms of either the coefficient of friction or wear rate by adding CuO to the polyblends with 10 wt.% PTFE.
4. GF reinforcement neither reduced wear nor the coefficient of friction of the polyblend.
5. Of all the methods used to improve the tribological behavior of the polyblends, 10% PTFE as the filler was the most effective. It gave a very low coefficient of friction of 0.17, which is exceptionally low for dry sliding at 1 ms^{-1} , and a low wear rate of $0.13 \times 10^{-6} \text{ mm}^3/\text{Nm}$.

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